

By Direction 1 of Problem 44, since $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j)$ is True, almost one of $P_1, P_2, \dots, P_n$ is true.

Case 1: $\forall i \in N, 1 \leq i \leq n, P_i$ is False.

By defn. of OR, $(P_1 \vee P_2 \vee \dots \vee P_i \vee \dots \vee P_n)$ is False.

This is a contradiction.

Case 2: Exactly one of $P_1, P_2, \dots, P_n$ is True. The rest are False.

By defn. of OR, $(P_1 \vee P_2 \vee \dots \vee P_i \vee \dots \vee P_n)$ is True.

No contradiction

Direction 2: Exactly one of the propositions $P_1, P_2, \dots, P_n$ is True $\Rightarrow$

$\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \wedge (P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)$ is True.

By Direction 2 of Problem 44, $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j)$ is True.

By Defn. of OR, $(P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)$ is True.

By Defn. of AND, $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \wedge (P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)$ is True.

Fuzzy logic is used in artificial intelligence. In fuzzy logic, a proposition has a truth value that is a number between 0 and 1, inclusive. A proposition with a truth value of 0 is false and one with a truth value of 1 is true. Truth values that are between 0 and 1 indicate varying degrees of truth. For instance, the truth value 0.8 can be assigned to the statement "Fred is happy," because Fred is happy most of the time, and the truth value 0.4 can be assigned to the statement "John is happy," because John is happy slightly less than half the time. Use these truth values to solve Exercises 49-51

49) The truth value of the negation of a proposition in fuzzy logic is 1 minus the truth value of the proposition. What are the truth values of the statements "Fred is not happy" and "John is not happy"?

Soln: Fred is not happy = $1 - 0.8 = 0.2$
John is not happy = $1 - 0.4 = 0.6$